

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

$$\sum x_i = 5, \bar{x} = 1, \sum y_i = 10, \bar{y} = 2$$

a.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i =$ 5	$\sum y_i =$ 10	$\sum (x_i - \bar{x}) =$ 0	$\sum (x_i - \bar{x})^2 =$ 10	$\sum (y_i - \bar{y}) =$ 0	$\sum (x_i - \bar{x})(y_i - \bar{y}) =$ 8

b. $b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = 0.8$, x 每±1单位, y 平均±增加0.8单位

$b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 = 1.2$, $x=0$ 时, y 平均为 1.2

c. $\sum x_i^2 = 9 + 4 + 1 + 1 + 0 = 15$

$\sum x_i y_i = 12 + 4 + 3 + (-1) = 18$

$\sum x_i^2 - N \bar{x}^2 = 15 - 5 \times 1^2 = 10 = \sum (x_i - \bar{x})^2$

$\sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \times 1 \times 2 = 8 = \sum (x_i - \bar{x})(y_i - \bar{y})$

d. $\hat{y}_i = 1.2 + 0.8 x_i$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i =$ 5	$\sum y_i =$ 10	$\sum \hat{y}_i =$ 10	$\sum \hat{e}_i =$ 0	$\sum \hat{e}_i^2 =$ 3.6	$\sum x_i \hat{e}_i =$ 0

$s_{yy} = \frac{\sum (y_i - \bar{y})^2}{n-1} = 2.5$

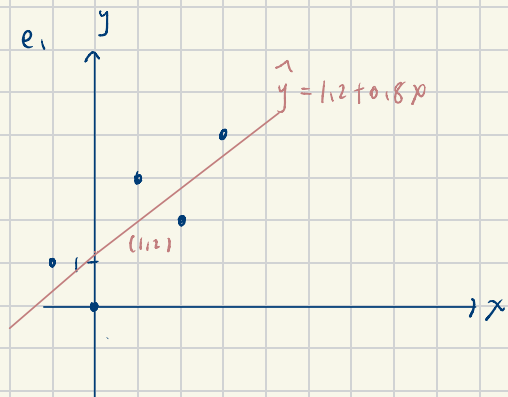
$s_{xx} = \frac{\sum (x_i - \bar{x})^2}{n-1} = 2.5$

$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1} = 2$

$r_{xy} = \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}} = \frac{2}{2.5} = 0.8$

$CV_x = 100 \times \frac{\sqrt{\frac{s}{2}}}{1} = 158.11\%$

$x_{me} = 1$



f. the line must pass through $(\bar{x}, \bar{y})$

g.

$$b_1 + b_2 \bar{x} = 1.2 + 0.8 \times 1 = 2 = \bar{y}$$

h.

$$\bar{\hat{y}} = \frac{10}{5} = 2 = \bar{y}$$

i.

$$\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$$

j.

$$\widehat{\text{Var}}(b_2 | X) = \frac{\hat{\sigma}^2}{\sum (X_i - \bar{X})^2} = \frac{1.2}{10} = 0.12$$

$$SE(b_2) = \sqrt{0.12} = 0.3464$$

2.22

a.

```
Call:
lm(formula = totalscore ~ small, data = data_a)

Residuals:
    Min       1Q   Median       3Q      Max
-189.44  -50.44   -9.44   43.06   324.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  916.442      3.675  249.396  <2e-16 ***
small         12.175      5.369   2.268  0.0236 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared:  0.006608, Adjusted R-squared:  0.005323
F-statistic: 5.142 on 1 and 773 DF, p-value: 0.02363
```

b_2 在 95% 的置信区间下显著

且小工班表现较好。

平均而言多 12.175 分

b.

```
Call:
lm(formula = readscore ~ small, data = data_a)

Residuals:
    Min       1Q   Median       3Q      Max
-74.665 -21.590  -2.665  16.410  194.335

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665      1.488  290.760  < 2e-16 ***
small         6.924      2.174   3.185  0.00151 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295, Adjusted R-squared:  0.01167
F-statistic: 10.14 on 1 and 773 DF, p-value: 0.001507
```

small 对 readscore 平均比 small=0 多 6.924 分

(99% 的置信区间下)

```
Call:
lm(formula = mathscore ~ small, data = data_a)

Residuals:
    Min       1Q   Median       3Q      Max
-144.777  -35.028   -5.028   29.223  142.223

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777      2.449  197.545  <2e-16 ***
small         5.251       3.578    1.467    0.143
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778, Adjusted R-squared:  0.001488
F-statistic: 2.153 on 1 and 773 DF, p-value: 0.1427
```

在 mathscore 上
結果不顯著。

```
Call:
lm(formula = totalscore ~ aide, data = data_c)

Residuals:
    Min       1Q   Median       3Q      Max
-191.748  -50.442   -5.442   40.558  312.558

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  916.442      3.559  257.529  <2e-16 ***
aide         4.306       4.994    0.862    0.389
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 72.23 on 835 degrees of freedom
Multiple R-squared:  0.0008898, Adjusted R-squared: -0.0003068
F-statistic: 0.7436 on 1 and 835 DF, p-value: 0.3888
```

無明顯證據證明
aide 對 totalscore 有影響

```
Call:
lm(formula = readscore ~ aide, data = data_c)

Residuals:
    Min       1Q   Median       3Q      Max
-74.665  -21.536   -2.536   15.464   194.335

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665      1.460  296.341  <2e-16 ***
aide         2.871       2.049    1.401    0.161
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.64 on 835 degrees of freedom
Multiple R-squared:  0.002347, Adjusted R-squared:  0.001152
F-statistic: 1.964 on 1 and 835 DF, p-value: 0.1615
```

```
Call:
lm(formula = mathscore ~ aide, data = data_c)

Residuals:
    Min       1Q   Median       3Q      Max
-146.212  -31.212   -5.777   29.223  142.223

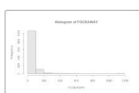
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777      2.355  205.464  <2e-16 ***
aide         1.435       3.304    0.434    0.664
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 47.79 on 835 degrees of freedom
Multiple R-squared:  0.0002258, Adjusted R-squared: -0.0009715
F-statistic: 0.1886 on 1 and 835 DF, p-value: 0.6642
```

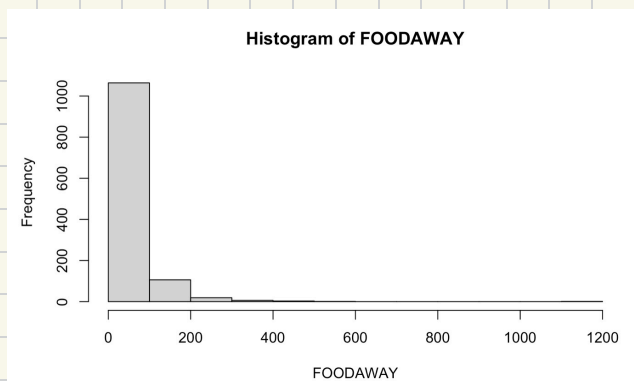
在 Readsore 和 mathscore; aide 對結果無顯著影響

```
## {r}
# 2.25(a)
summary(cex5_small$foodaway)
hist(cex5_small$foodaway, main = "Histogram of FOODAWAY",
      xlab = "FOODAWAY")
```

R Console



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	12.04	32.55	49.27	67.50	1179.00



b、

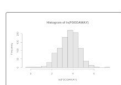
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
0.00	21.67	48.15	73.15	90.00	1179.00	Advanced degree
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
0.00	14.44	36.11	48.60	68.67	416.11	College degree
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
0.00	9.63	26.02	39.01	52.65	437.78	No degree

c、

```

{r}
# 2.25(c)
summary(log(cex5_small$foodaway[cex5_small$foodaway > 0]))
hist(log(cex5_small$foodaway[cex5_small$foodaway > 0]),
     main = "Histogram of ln(FOODAWAY)",
     xlab = "ln(FOODAWAY)")

```

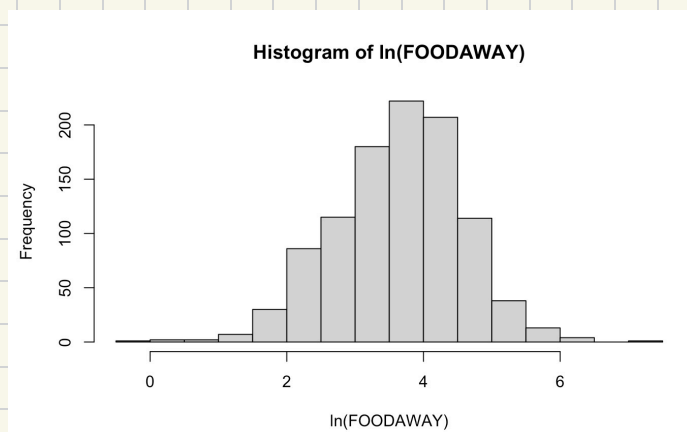


Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.3011	3.0759	3.6865	3.6508	4.2797	7.0724

1. 对变量取对数时，会异常。

foodaway = 0 的样本是剔除

故样本数量下降



d.

```
Call:
lm(formula = log(foodaway) ~ income, data = cex5_sub)

Residuals:
    Min       1Q   Median       3Q      Max
-3.6547 -0.5777  0.0530  0.5937  2.7000

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.1293004   0.0565503   55.34  <2e-16 ***
income       0.0069017   0.0006546   10.54  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

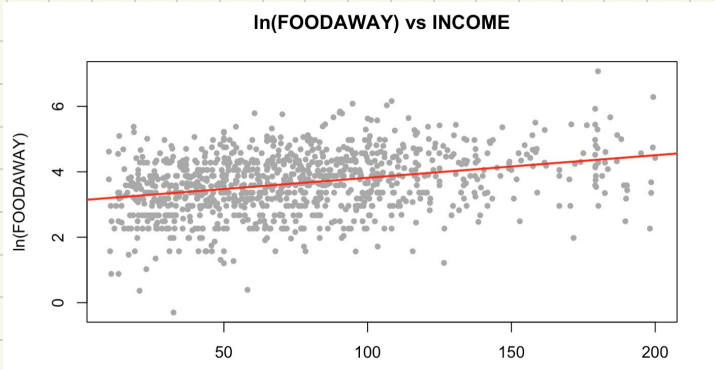
Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826, Adjusted R-squared:  0.09738
F-statistic: 111.1 on 1 and 1020 DF, p-value: < 2.2e-16
```

$$\ln(\text{foodaway}) = 3.1293 + 0.0069 \times \text{income}$$

income 增加 1% 則

foodaway 平均而言增加 $3.1293 + 0.0069 \times 100 = 0.7\%$

e.



可能存在 heteroscedasticity 問題

f.

